

Microscopic implications of S-DNA

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Abstract

Recent experiments [J. van Mameren *et al.* PNAS 106, 18231 (2009)] provide a detailed spatial picture of overstretched DNA, showing that under certain conditions the two strands of the double helix separate at about 65 pN. It was proposed that this observation rules out the existence of an elongated, hybridized form of DNA (S-DNA). Here we argue that the S-DNA picture *is* consistent with the observation of unpeeling during overstretching. We demonstrate that assuming the existence of S-DNA does not imply DNA overstretching to consist of the complete or near-complete conversion of the molecule from B- to S-form. Instead, this assumption implies in general a more complex dynamic coexistence of hybridized and unhybridized forms of DNA. We argue that such coexistence can rationalize several recent experimental observations.

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